
CAT DILR

Full-Corpus PYQ

Inference Report

Reverse-engineering the reasoning architecture of CAT DILR
across nine years (2017–2025), and a build specification
for a CAT-grade DILR generator

Prepared for ExamIQ Platform — DILR Generation Initiative

Corpus CAT PYQ.pdf (2017–2025, 693 pp)
+ CAT 2025 Slot 1/2/3 solved papers

Coverage 24 DILR slot-sections, ~129 DILR sets, 2017–2025

Status Inference & specification document
No practice questions, mocks, or new sets are generated

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1. Executive Summary

This report rebuilds the previous DILR inference, which analysed only CAT 2025 (three slots, 15 sets, one year). **This version analyses the full historical corpus — ~129 DILR sets spanning CAT 2017 through 2025** — with CAT PYQ.pdf as the primary expansion and the three 2025 solved papers providing solution-grade depth.

The expansion reinforces the earlier central thesis: **a CAT DILR set is a hidden object the student must reconstruct, not information to be read**. But several secondary conclusions were **overfit to 2025**:

1. **“Venn / set-theory sets are gone”** — false. Venn is the **6th most common family (12 sets, ~9%)**, a per-slot staple in 2018, 2020 and 2022; it merely sat out 2021, 2023 and 2025.
2. **“Tournaments don’t appear”** — false. **Tournament / score-reconstruction is a recurring family (8 sets)**: Hi-Lo bidding, betting-pairs, javelin, archery, rifle, the Chango knockout, the QUIET schedule, and 2025’s promote-demote set.
3. **“Calculation is never the whole step”** — overstated. The corpus has a real vein of **near-pure quant-DI** (2023 broadly; 2024 GDP/candlestick/bubble). **quant-DI is the single most common surface family (22 sets, ~17%)**.
4. An entire engine class — **number-theory / cipher reconstruction** (cryptarithms, letter→digit ciphers, prime-product allocation) — recurs in 2018–2022 and was **invisible to the 2025-only view**.

The corpus also reveals a **structural era-shift**: 2017–2019 ran **8 sets / 32 questions per slot** with high arithmetic intensity and number-theory engines; 2020 was a transition; 2021–2025 settled at **4–5 sets / 20–22 questions per slot**, hybrid-first and deeper per set.

2. Source Coverage (summary)

Full detail is in `source_coverage_table.md`. All four mandatory PDFs were detected in `sources/cat_pyqs/`, are readable, carry real text layers (no OCR), and are included.

File	Pages	DILR	Years	Role
CAT PYQ.pdf	693	24 sections	2017–2025	Primary corpus (~129 sets)
2025 Slot-1 Q-Sol	28	Section II	2025 S1	Solution-grade (5 sets)
2025 Slot-2 Q-Sol	28	Section II	2025 S2	Solution-grade (5 sets)
2025 Slot-3 Q-Sol	32	Section II	2025 S3	Solution-grade (5 sets)

Gate: PASSED — CAT PYQ.pdf present, readable, and the analytical backbone. Limitation: 2017–2024 are questions-only (no examiner solutions), so casework for those years is structurally inferred and tagged (*inf*); CAT 2025 solutions calibrate that inference.

3. Methodology

1. **Source gate first.** Inspected `sources/cat_pyqs/`, verified all four PDFs, confirmed text layers with PyMuPDF. No analysis before the gate passed.
2. **Structural extraction.** `pdftotext -layout` on every file; for CAT PYQ.pdf, year blocks and section headers located programmatically and the 24 DILR sections sliced per slot.

3. **Per-year classification.** Each section segmented into sets; each set scored on surface family, true engine, visible/hidden/inferable split, casework, a five-axis difficulty vector, ladder character, and CAT-likeness.
4. **Difficulty vector (L/R/C/D/M), 0–3:** **L** logic depth, **R** representation difficulty, **C** calculation load, **D** data density, **M** dynamic movement / feedback.
5. **Calibration.** CAT 2025 solutions anchored casework patterns; older-year casework inferred from constraint structure and necessity phrasing.
6. **Integrity.** No question text reproduced; invented labels and structural descriptions only.

4. Full-Corpus Findings

F1 — The hidden object is universal.

In every set a structure is fully specified in the examiner’s mind but only partially exposed; the work is reconstruction.

F2 — Hybrids dominate, increasingly.

The share of sets whose true engine is an explicit hybrid of two archetypes rises markedly from 2021.

F3 — Arithmetic is a real difficulty axis — sometimes the whole axis.

~40% of sets carry $C \geq 2$; a minority (2023; 2024 GDP/candlestick/bubble) are near-pure quant-DI.

F4 — Designed indeterminacy is pervasive.

Necessity / “cannot be determined” / range answers recur in the majority of sets in every year.

F5 — Family mix rotates year to year.

No family is guaranteed in a given year, but over a multi-year window all nine families recur. Tuning to one year misses families that sat out.

5. Set Family Taxonomy & Frequency

Classifying all ~129 sets by **surface family** (exact tallies from Appendix A):

Surface family	Sets	Share	Peak years	Absent in
quant-DI	22	17%	2023, 2024	—
matrix / grid	20	16%	2019, 2020, 2023	—
distribution / allocation	19	15%	2018, 2022, 2025	—
chart-plus-logic	19	15%	2019, 2024, 2025	—
scheduling	15	12%	2017, 2019	—
Venn / set-theory	12	9%	2018, 2020, 2022	2021, 2023, 2025
routes / network	9	7%	2017	2018, 2020
tournament	8	6%	2019, 2021	2017, 2018, 2023
dynamic-movement	3	2%	2025	most years
hybrid (surface)	2	2%	2017–2018	—

- The “big four” (quant-DI, matrix/grid, distribution, chart-plus-logic) are ~63% of all sets — the generator’s core output.

- **Venn and tournament are not optional** ($\sim 15\%$ combined, multi-year cycle). The 2025-only report wrongly retired both because both sat out 2025.
- **dynamic-movement is rare but high-signal** (M=3 sets): a premium, occasional output, not a staple.

5.1 The structural era-shift

Era	Sets/slot	Qs/slot	Character
2017–2019	8	32	High arithmetic; number-theory engines (ciphers, cryptarithms); heavy networks (airline, congestion/Braess); dense necessity culture
2020	6	~ 24	Transition; logic flagships (binary-encoding blood-vial, parking simulation); strong Venn
2021–2025	4–5	20–22	Hybrid-first; fewer but deeper sets; reconstruction fused with arithmetic; movement re-emerges in 2025

Target the **modern shape** (4–5 deep hybrid sets/slot) with the **full-corpus vocabulary**.

6. The Hidden Reasoning Engine

The ten surface families collapse into a smaller set of true-engine archetypes.

Engine archetype	Reconstructed object	Representative sets (year)
Grid / matrix	2-D table of relationships	platform-height ('17), pouch coins ('19), bead grid ('20), call-ledger ('25)
Timeline / sequence	who-does-what-when blocks	dancers ('19), seminars ('20), journals ('21), Gurus ('25)
Allocation	distribution of discrete units	inheritance ('17), committees ('18), envy-free ('21), train occupancy ('25)
Ranking / score table	ordering, often time-varying	tea-taster ('17), Hi-Lo ('20), promote-demote ('25)
Movement system	trajectory through states	parking ('20), patrol ('23), passing-buck & round-table ('25)
Flow network	conserved directed volumes	elective flow ('17), town friendship ('21), trade flows ('25)
Venn region-set	counts in overlaps, often ranges	CET ('17), satellites ('18), medicine ('20), travelers ('24)
Reconstruct-from-rates	absolutes from ratios/%	smartphones ('18), store doubling ('19), SI scatter ('25)
Number-theory / cipher	code/factorisation forced by arithmetic	cryptarithm ('19), letter cipher ('18), prime funding ('22)

Key insight: the hardest sets are **hybrids of two archetypes** (train occupancy = allocation + flow network; promote-demote = ranking + feedback; prime funding = allocation + number-

theory). A generator that builds only single-archetype objects sits a full difficulty tier below CAT.

7. Visible vs Hidden vs Inferable Information

A deliberate three-layer design, consistent across all nine years:

1. **Visible** — scaffolding: entities, rule, units, a chart, a few anchor facts.
2. **Hidden** — the body of the object: most cells, positions, assignments, volumes.
3. **Inferable-only** — facts that appear nowhere and emerge only after combining clues; the target of hard/global questions.

Set (year)	Visible	Hidden	Inferable-only
CET percentiles ('17)	“exactly two” rule, a ratio, one count	3-set Venn region sizes	Triple-overlap is a <i>free variable</i> $\Rightarrow$ range answer
Tea-taster ('17)	doubling/even/min clues	cup $\leftrightarrow$ origin $\leftrightarrow$ rating bijection	Which statements are only “ <i>may be true</i> ”
Parking ('20)	arrival/exit rule, slot map	car type & occupancy over time	State at a queried moment, per scenario
Promote-demote ('25)	leaders, range, tie-break, ~ 4 ratings	full rating grid + scores + membership	Whose score is <i>undeterminable</i> at a quarter
Train occupancy ('25)	capacity, few %, multiples-of-10	full origin–destination matrix	Counts only the full matrix pins

Design principle: anchor facts must under-determine the object alone and become sufficient only in combination; no single fact resolves a cell directly. The “NOT determinable / NOT necessarily true” question type (every year) is the explicit test of layer 3.

8. Real Casework Analysis

Casework = the deduction forks, ≥ 2 live hypotheses are carried, and a **non-local** later clue kills all but one.

- **Heavy, genuine casework is the norm** in every year; exceptions cluster in near-pure quant-DI (read-and-compute).
- **The killing clue is non-local** — different condition or later question premise (2025 solutions show this explicitly: the $x=7$ tap branch dies on an impossibility three steps away; the $(x+60)=96$ SI branch dies on a lowest-rank condition stated elsewhere).
- **At least one branch is seductive** — arithmetically clean, dying only under a divisibility/parity/range test. This is why **over-clean arithmetic destroys realism**: nothing for the false branch to die on.
- **Designed residual ambiguity** — many sets are correct but not fully solvable (Venn ranges; “cannot be determined”; “may be true”).

Family	Typical casework
matrix/grid, allocation, scheduling, tournament, movement, Venn	Heavy — forced forks, non-local kills, often residual ambiguity
reconstruct-from-rates, number-theory	Medium-heavy — divisibility/factorisation forks
quant-DI (pure), chart read-off	Light / none — the arithmetic is the point

9. Difficulty Driver Analysis

Driver	Corpus tendency
L — logic depth	High in grid/allocation/scheduling/tournament/movement/number-theory; low in pure quant-DI
R — representation	Consistently high (≥ 2) in most sets — choosing how to draw the object is half the battle; the most under-rated driver
C — calculation	Bimodal — spikes to 3 in quant-DI / reconstruct-from-rates / older arithmetic sets; near 0 in logic flagships
D — data density	Moderate-high in DI/chart sets; high D without high L = textbook DI, not CAT
M — movement / feedback	Rare but decisive (parking '20, passing-buck/round-table '25); hardest to fake

The stacking law: CAT-level difficulty comes from stacking ≥ 2 drivers, not cranking one. High on C alone = a calculation drill; high on D alone = textbook DI. Neither is CAT-level.

10. The Question Ladder

Rung	Purpose	Typical form
Entry	Reward the first solid deduction	A value that falls out early
Medium	Require a partially-built object	A value needing 2–3 chained clues
Hard	Require the fully reconstructed object	A value/comparison resolving only at the end
Global / conditional / necessity	Test command of the whole structure or robustness	“For how many... determinable?”, “NECESSARILY true / CANNOT be true”, “If X then...”

The necessity / “cannot be determined” rung is present in the majority of sets every year (dominant 2017–2020; central 2024–2025). Pure quant-DI sets are the exception and the easiest

family. **Author a graded ladder with at least one global/necessity question; where the object is under-determined, at least one question must test determinacy.**

11. What Changed After Including CAT PYQ.pdf

11.1 Conclusions that remain valid

The hidden-object thesis; hybrid-first; representation as the under-rated driver; designed indeterminacy; the stacking law; “never hand over the graph” (confirmed by historical network sets: elective flow ’17, town friendship ’21, dean votes ’23).

11.2 Conclusions that were overfit to 2025

2025-only claim	Full-corpus correction
“Venn sets gone $\Rightarrow$ de-prioritise.”	Venn is the 6th-most-common family (12 sets); staple in 2018/2020/2022; only sat out 2021/2023/2025. Keep Venn.
“Tournaments don’t appear.”	Tournament/score-reconstruction recurs (8 sets): Hi-Lo, betting, javelin, archery, rifle, Chango, QUIET, promote-demote. First-class family.
“Calculation is never the whole step.”	Near-pure quant-DI is a real vein (2023; 2024 GDP/candlestick/bubble). quant-DI is the most common surface family (22). Produce a controlled minority of calc-dominant sets.
(silent) “ $\sim 4\text{--}5$ sets/slot is the norm.”	True now, but 2017–2019 ran 8 sets / 32 Q per slot — a larger engine vocabulary.

11.3 Historical patterns missing / under-weighted in the 2025-only view

- **Number-theory / cipher** engines (cryptarithm ’19, letter cipher ’18, prime funding ’22) — an entire class invisible to 2025-only.
- **Equilibrium / congestion networks** (2017 four-car Braess-style, airline connectivity) — game-theoretic flow networks.
- **Simulation / state-machine** sets (parking ’20, project-team rotation ’17) — pure dynamic-state engines.
- **Extremal / min-max optimisation** (100-boxes ’19, ATM notes ’18, cab ratings ’20).

11.4 How the generation strategy should change

1. Sample families from the full-corpus distribution, not 2025 — reinstate Venn, tournament, number-theory, simulation, equilibrium-network.
2. Target the modern *shape* (4–5 deep hybrid sets/slot) with the historical *vocabulary*.
3. Add a controlled minority of calculation-dominant quant-DI sets to match CAT’s real mix.
4. Make necessity / determinacy questions mandatory.

11.5 Families to prioritise

Tier 1 (core, every batch): matrix/grid, distribution/allocation, scheduling/timeline, quant-DI (incl. reconstruct-from-rates).

Tier 2 (regular rotation): chart-plus-logic, Venn/set-theory, tournament/score-reconstruction, flow-network.

Tier 3 (premium, occasional): dynamic-movement/simulation, number-theory/cipher, equilibrium-network.

12. Why Our AI-Generated DILR Failed

#	Failure mode	Practical rejection signal
1	Direct graph/structure reading	A question is answered by quoting the diagram
2	Graph/table handed over	The set ships with the grid/graph drawn
3	Small universe	Hand-solvable in < 60 s
4	Fake casework	Every “case” closes in the next clue
5	One-step deductions	Each clue resolves exactly one cell
6	Over-clean arithmetic	No integer/parity/multiple/range constraint eliminates
7	No hidden object	Cannot name a structure to build that isn’t printed
8	No dynamic state	Nothing changes as you deduce (when theme implies it should)
9	No residual ambiguity	One clean world; no honest “cannot be determined”
10	Weak question ladder	All questions same difficulty
11	No global/conditional/necessity question	No “necessarily true”/“if X then”
12	Answerable directly from given facts	The question restates a printed fact
13	Uniqueness verified but not CAT-likeness	Solver confirms one answer, but the set is step/clean/static/ladderless

The deepest failure is #13. Our prior pipeline verified *answer uniqueness* and stopped — but a set can be unique and still trivial. **Uniqueness is necessary, not sufficient.** The generator needs a second gate that checks *CAT-likeness* (multi-driver difficulty, forced non-local casework, a hidden object, a proper ladder), not just solvability.

On the graph mock: it fails #1, #2, #3, #7, #10 at once. The fix is not “avoid networks” but **“never hand over the network”** — CAT’s network sets (elective flow ’17, town friendship ’21, dean votes ’23, trade flows ’25) reconstruct the network from balances/conservation/aggregates, never drawn.

13. Full-Corpus Generation Rules

- Object-first authoring.** Build the complete hidden object first, then derive clues by deletion + indirection. Never generate clues first.
- Hidden representation.** Do not hand over the final graph/table/grid; if a diagram is shown, the challenge must be reconstructing a further hidden layer behind it.
- Forced casework.** ≥ 2 globally-consistent partial worlds survive past the initial clue reading.

4. **Non-local clue interaction.** The clue that kills a false case must not be adjacent to the clue that created the fork.
5. **Multi-driver difficulty (stacking law).** Score ≥ 2 on at least two of {L,R,C,D,M}. C-alone or D-alone is rejected.
6. **Proper question ladder.** Entry + medium + hard + at least one global/conditional/necessity; test determinacy where the object is under-determined.
7. **Arithmetic that does logical work.** Numbers carry constraints (integer, parity, multiples, distinctness, range) so a clean-but-wrong branch has something to die on.
8. **Solver-verified determinacy.** A formal solver / enumerator confirms, per question, unique vs intentionally indeterminate.
9. **AI quality gate (CAT-likeness, not just uniqueness).** A separate judge tries to reject on the checklist, specifically catching “unique but trivial” (#13).
10. **Family/engine sampling from the full corpus.** Draw from the §6 distribution (incl. Venn, tournament, number-theory, simulation); target the modern 4–5-set shape.

14. Rejection Checklist

Run on every generated set before PDF rendering. A single tick is an automatic reject.

- ✗ Any answer is **directly readable** from a clue, chart, or graph.
- ✗ **One early deduction collapses** the whole set.
- ✗ **No hidden object** must be reconstructed.
- ✗ The final **graph/table/grid is handed over** (no further hidden layer).
- ✗ **No real casework** survives past the first clue reading.
- ✗ The **universe is too small** ($< \sim 60$ s by hand).
- ✗ **Arithmetic is the only difficulty** (C high, L/R/M all low).
- ✗ **All questions are the same difficulty** (no ladder).
- ✗ **No question tests global structure.**
- ✗ **No question tests conditional or necessity** reasoning.
- ✗ **No question tests determinacy** where the object is intentionally under-determined.
- ✗ The set is **merely chart reading + calculation.**
- ✗ The **blind solver solves it too quickly, with no branching.**
- ✗ The **formal solver cannot verify** uniqueness or intended ambiguity.
- ✗ (*meta*) The set is **unique but not CAT-like** (passes uniqueness yet trips ≥ 3 above — failure #13).

Acceptance bar: no box ticked; rules §14.1–§14.10 hold; an independent blind solver reconstructs the object from the prompt alone and reproduces both the intended answers and the intended indeterminacies; the CAT-likeness judge returns accept.

15. Recommended Terminal Pipeline

Core principle: **separate the author from the checkers**, and gate on **CAT-likeness**, not just uniqueness.

1. SOURCE INGESTION Place PDFs in sources/cat_pyqs/. Verify text layer; pdftotext.
2. PYQ SET EXTRACTION Slice year/slot/section; segment sets; structural digests.
3. REASONING INFERENCE Classify family+engine+driver vector; corpus dist = sampler prior.
4. HIDDEN-OBJECT
AUTHORING Sample (family, engine-hybrid, driver target). Build FULL object first, then delete+re-express into clues. Author graded ladder (object-first) incl. a necessity/determinacy question.
5. BLIND SOLVING SEPARATE agent (no key) reconstructs object from clues; logs deduction order, branch count, time-to-solve.
6. FORMAL VERIFICATION z3 / OR-Tools / brute enumeration counts ALL worlds -> per-Q determinacy profile (unique vs intended-ambiguous).
7. ADVERSARIAL JUDGING SEPARATE judge runs rejection checklist incl. CAT-likeness gate. Any tick -> reject, loop to stage 4 with reasons.
8. LaTeX / PDF RENDER Template set.tex + solution.tex; pdflatex (x2 for ToC).
9. WEBSITE-READY OUT Emit set.json (prompt, options, answers, determinacy flags) + HTML.
10. STORAGE / INDEXING content/cat/dilr/<id>; register in practice DB; index by family, engine, driver vector, difficulty, world-count, ladder profile.

15.1 Proposed folder structure

```
content/cat/dilr/
  <set_id>/
    prompt.json      # entities, rule, anchor clues, questions, options
    answer_key.json  # full hidden object + per-question answers + determinacy
    metadata.json    # family, engine, drivers, world-count, judge verdict, provenance
    set.tex set.pdf
    solution.tex solution.pdf
    web.json         # website-ready payload
    _rejected/<set_id>/ # rejected drafts + structured failure reasons
sources/cat_pyqs/    # raw PYQ PDFs (never committed; analysis input only)
reports/dilr_inference_full/ # this report + coverage/extraction logs
```

15.2 Metadata to store per set

engine_archetype, engine_hybrid_of[], surface_family, driver_vector {L,R,C,D,M}, entity_count, consistent_world_count, casework_branch_count, has_indeterminate_question, ladder_profile, blind_solver_minutes, blind_solver_branches, judge_verdict, source_family_prior, provenance.

15.3 Orchestration & repo hygiene

One dilr_pipeline.py (or Claude Code workflow), -count N. Author / Solver / Judge are distinct agents; solver and judge run without the key. Formal verifier: **z3-solver** / **ortools** (or brute enumeration). Only stage-7-passing sets reach content/. **Never** commit/move raw PDFs; never touch app/source, next-env.d.ts, tsconfig.json, package/build config, or .next/.

16. Appendix A — Per-Year Set Inventory

Compact teardown of all ~129 sets. Drivers are L/R/C/D/M (0–3). Casework for 2017–2024 is (*inf*) = inferred from structure; 2025 is solution-calibrated. Labels are invented descriptors — no question text is reproduced.

16.1 CAT 2017 (16 sets — 8/slot; high arithmetic, networks, number-theory)

Set		Family	Engine	Casework	Drivers	Ladder
Pizzeria proportion	type-	quant-DI	matrix (party×type)	light	L1 R2 C3 D2 M0	partial
Elective enrollment flow	re-	routes/network	flow-network	heavy	L3 R3 C2 D3 M1	yes
Inheritance split	asset	distribution	allocation	heavy	L3 R1 C2 D1 M0	partial
Dorm grid	repair-cost	chart-plus-logic	grid	heavy	L3 R2 C1 D2 M0	yes
Tea-taster ratings	cup rat-	scheduling	ranking	heavy	L3 R2 C1 D1 M0	yes
Queen board	attack chess-	matrix/grid	grid geometry	light	L2 R3 C2 D1 M1	partial
Airline seat pricing		scheduling	allocation	heavy	L3 R2 C3 D3 M0	yes
Fingerprint order count	scan-	hybrid	sequence (perm.)	light	L3 R1 C2 D0 M0	yes
Fast-food scheduling	order	scheduling	timeline (parallel)	heavy	L2 R2 C2 D2 M2	yes
Rural-kids survey	school	quant-DI	3-way matrix	light-heavy	L2 R2 C3 D3 M0	partial
CET percentile selection		Venn	Venn-regions	heavy	L3 R2 C2 D2 M0	yes (range)
Happiness-index scores		chart-plus-logic	grid (S/F/C)	heavy	L3 R2 C2 D2 M0	yes
Project-team rotation	SE/RE	distribution	movement-traj.	heavy	L3 R2 C3 D2 M3	yes
Platform-height reachability		matrix/grid	grid (sightline)	heavy	L3 R3 C1 D2 M0	yes
Airline network connectivity		routes/network	flow-network	heavy	L3 R2 C2 D1 M0	yes
Four-car route equilibrium	route equi-	routes/network	flow-network (eq.)	heavy	L3 R1 C3 D2 M2	yes (Braess)

16.2 CAT 2018 (16 sets — 8/slot; logic-heavy, Venn ×2, ciphers)

Set		Family	Engine	Casework	Drivers	Ladder
Currency buy/sell inventory		quant-DI	score-table	light	L1 R1 C3 D2 M1	partial
Letter→digit cipher		number-theory	allocation+deduction	heavy	L3 R2 C1 D2 M0	yes
Smartphone share/profit		quant-DI	score-table	none	L0 R1 C3 D1 M0	no
Interview-room arrival log		scheduling	timeline	heavy	L3 R2 C1 D1 M1	yes
Accreditation weights		matrix/grid	score-table (perm.)	heavy	L3 R2 C3 D2 M0	yes
Product grid	portfolio	chart-plus-logic	matrix	heavy	L3 R2 C2 D2 M0	partial
Amusement-park tickets		distribution	matrix (age×ticket)	heavy	L2 R2 C3 D3 M0	yes

Set		Family	Engine	Casework	Drivers	Ladder
Sports Venn	enrollment	Venn	Venn-regions	heavy	L2 R2 C3 D2 M0	yes (min/max)
$n \times n$ numeral colouring		matrix/grid	grid (king-graph)	light	L3 R0 C2 D1 M0	partial
Recruitment composite scores	com-	quant-DI	score-table	heavy	L3 R2 C3 D2 M0	yes
Multi-layer pie sales		chart-plus-logic	score-table (nested)	light	L1 R1 C3 D2 M1	no
Petrol-pump contamination	con-	scheduling	sequence (positional)	heavy	L3 R2 C2 D2 M0	yes
Satellite purposes		Venn	Venn-regions	heavy	L2 R2 C3 D2 M0	yes (range)
ATM dispensing	note-	distribution	allocation (partition)	heavy	L2 R2 C3 D2 M0	partial
Three-committee composition		distribution	matrix	heavy	L3 R2 C3 D2 M0	yes
Students jor/minor	ma-	matrix/grid	grid (triple-attr.)	heavy	L3 R2 C1 D1 M0	yes

16.3 CAT 2019 (16 sets — 8/slot; sequence/grid, cryptarithm, tournaments)

Set		Family	Engine	Casework	Drivers	Ladder
Store shares	revenue/cost	quant-DI	score-table	heavy	L1 R1 C2 D3 M0	partial
Pouch grid	min-max coins	matrix/grid	grid (integer-avg)	heavy	L3 R1 C3 D2 M0	partial
Two-proposal survey	voter	Venn	Venn-regions	heavy	L2 R1 C3 D2 M0	partial
Speakers/teams/language		matrix/grid	grid+allocation	heavy	L3 R2 C3 D1 M0	partial
Doctors rooms/queue tokens	to-	scheduling	timeline (queue)	heavy	L2 R1 C2 D2 M2	partial
Monsoon rainfall LPA	vs	chart-plus-logic	score-table (roll.)	light	L1 R1 C2 D1 M0	partial
Faculty question order	MT/ET	scheduling	sequence	heavy	L3 R2 C3 D1 M0	yes
Rifle shooting rounds	10	tournament	score-table (window)	heavy	L3 R3 C3 D2 M0	partial
Cryptarithm 6-digit		number-theory	grid+ranking (carry)	heavy	L3 R1 C3 D2 M0	partial
Dancers/composers order		scheduling	sequence	heavy	L3 R2 C3 D0 M0	partial
100-boxes prize doubling		distribution	allocation (extremal)	heavy	L3 R1 C2 D1 M0	yes
Supermarket shelves	16	distribution	sequence (cluster)	heavy	L3 R2 C3 D0 M0	yes
Street-map sight	line-of-	routes/network	movement-traj.	heavy	L3 R2 C3 D0 M2	partial
Archery rounds	bonus	tournament	score-table	heavy	L2 R3 C3 D2 M0	partial
Foreigner-crime ranking DI		chart-plus-logic	score-table+rank	light-med	L1 R3 C2 D1 M0	partial

Set		Family	Engine	Casework	Drivers	Ladder
Five-vendor scoring	radar	chart-plus-logic	score-table	none-light	L1 R1 C1 D2 M0	no

16.4 CAT 2020 (15 sets — 5/slot; logic flagships, strong Venn, simulation)

Set		Family	Engine	Casework	Drivers	Ladder
Pooled blood-vial diagnosis		matrix/grid	matrix+binary-encoding	heavy	L3 R2 C2 D0 M0	yes
Hi-Lo bidding	four-player	tournament	score-table	heavy	L3 R2 C2 D2 M0	yes
OTLP 4-facility survey		Venn	Venn-regions (4-set)	light	L1 R1 C1 D3 M0	partial
Two-day grocery delivery		quant-DI	timeline+score-table	light	L1 R2 C0 D3 M0	partial
Farmer's 12-plot land		matrix/grid	grid+allocation	heavy	L3 R2 C2 D2 M1	yes
5 × 5 tri-colour beads		matrix/grid	matrix+combinatorics	heavy	L3 R2 C3 D0 M1	partial
Doctoral scheduling	seminar	scheduling	timeline	heavy	L3 R2 C1 D0 M0	yes
Constituency deposit votes		chart-plus-logic	score-table+quant-DI	light-heavy	L2 R2 C1 D3 M0	partial
Departmental-store sales		quant-DI	matrix (AP fill)	light	L1 R2 C0 D3 M0	partial
Basement compact/SUV parking	com-	dynamic-movement	movement-traj.	heavy	L3 R2 C1 D0 M3	no
Four-medicine groups		Venn	Venn-regions (4-set)	light	L1 R1 C1 D3 M0	partial
Vendor-institute contracts		scheduling	timeline	heavy	L3 R2 C1 D1 M0	yes
Board-exam marks	missed	chart-plus-logic	score-table+matrix	heavy	L3 R2 C1 D2 M0	yes
Cab-driver ratings		quant-DI	score-table+allocation	heavy	L2 R2 C2 D2 M0	partial
Ten percussion experts	ex-	Venn	Venn-regions+logic	heavy	L3 R2 C1 D0 M0	yes

16.5 CAT 2021 (12 sets — 4/slot; hybrid-first, arithmetic-laced)

Set		Family	Engine	Casework	Drivers	Ladder
Javelin ranking	throw re-	tournament	score-table+rank+timeline	heavy	L2 R3 C2 D3 M1	yes
Reviewer routing	approval	distribution	flow-network+grid	heavy	L3 R3 C3 D1 M1	partial
Employee Gantt	project	chart-plus-logic	timeline+score-table	light	L1 R2 C1 D3 M1	no
Impurity bottle testing	bottle test-	quant-DI	allocation+flow-network	heavy	L2 R2 C3 D2 M0	partial
Order Gantt	shipping	chart-plus-logic	timeline	light	L1 R2 C1 D3 M1	no

Set		Family	Engine	Casework	Drivers	Ladder
Chango bracket	knockout	tournament	ranking+flow-network	heavy	L3 R2 C2 D2 M0	yes
Delivery bonus	ratings &	matrix/grid	grid+score-table	heavy	L3 R2 C3 D2 M1	partial
Envy-free bundles	object	distribution	allocation+score-table	heavy	L3 R3 C3 D2 M0	yes
Subcategory bars	sales	quant-DI	score-table	none-light	L0 R1 C1 D3 M0	no
Smoothie econ.	production	quant-DI	allocation+score-table	heavy	L2 R2 C3 D2 M1	partial
Journal scheduling	paper	scheduling	grid+timeline	heavy	L3 R3 C3 D2 M1	yes
Town work	friendship net-work	routes/network	Venn+flow-network	heavy	L3 R2 C3 D1 M0	yes

16.6 CAT 2022 (12 sets — 4/slot; Venn $\times 3$, prime-number allocation)

Set		Family	Engine	Casework	Drivers	Ladder
CS/non-CS grades	course	Venn	Venn-regions	light	L2 R2 C3 D1 M0	yes
Cohort age-bands	mortality	quant-DI	timeline	light	L1 R1 C3 D3 M0	partial
Pandemic daily new-cases		distribution	grid (bounded int.)	heavy	L3 R3 C2 D1 M0	yes
Betting-pairs tracker	money	tournament	score-table	heavy	L3 R2 C2 D1 M2	partial
Supermarket approvals	ap-	Venn	Venn-regions (2 $\times$ 2 $\times$ 2)	light	L2 R2 C3 D1 M0	yes
Widget supply routing		routes/network	flow-network	heavy	L2 R2 C2 D3 M2	partial
Four-company revenue/cost	rev-	chart-plus-logic	score-table	none	L1 R1 C0 D3 M2	no
TRICCEK door-to-door sales		distribution	grid	heavy	L3 R2 C2 D2 M1	partial
Four goals/matches	players	matrix/grid	grid (scoring)	heavy	L3 R3 C2 D1 M0	yes
Prime-token funding		number-theory	matrix (prime-fact.)	heavy	L3 R3 C2 D1 M1	yes
Attendance tiers	interest	Venn	Venn-regions (nested)	heavy	L2 R2 C3 D1 M0	yes
Metro line scheduling		scheduling	timeline (map)	light	L2 R2 C1 D1 M3	partial

16.7 CAT 2023 (12 sets — 4/slot; very high arithmetic, ZERO Venn)

Set		Family	Engine	Casework	Drivers	Ladder
AC dealer sales	variant	distribution	grid (4 $\times$ 2 $\times$ 2)	light	L1 R1 C1 D3 M3	partial
Course scoring	project-test	matrix/grid	score-table+ranking	heavy	L2 R2 C2 D2 M2	yes

Set	Family	Engine	Casework	Drivers	Ladder
Patrol teams street routes	routes/network	movement-traj.+timeline	heavy	L2 R2 C3 D0 M0	yes
Online vs offline registr.	quant-DI	score-table (min/max/med)	heavy	L1 R1 C3 D3 M1	partial
Firms funding rise-fall	quant-DI	sequence	heavy	L1 R2 C3 D3 M0	yes
Draw competition ranks	quant-DI	score-table+ranking	heavy	L1 R2 C3 D3 M0	yes
3 × 3 coin-sack boxes	matrix/grid	matrix (sums+medians)	heavy	L1 R1 C3 D3 M0	yes
Theme-park rides timing	scheduling	timeline+allocation	heavy	L2 R1 C3 D2 M2	yes
Dean election voting	distribution	flow-network+allocation	heavy	L2 R3 C3 D1 M0	yes
Restaurant worker ratings	quant-DI	score-table (stats)	heavy	L1 R1 C3 D3 M0	partial
Visa office slot processing	scheduling	timeline+flow-network	heavy	L2 R1 C2 D3 M1	yes
Housing-complex pricing	matrix/grid	grid+score-table	heavy	L2 R2 C3 D2 M1	yes

16.8 CAT 2024 (15 sets — 5/slot; chart-plus-logic heavy, Venn returns)

Set	Family	Engine	Casework	Drivers	Ladder
ATM cash on road grid	routes/network	matrix+allocation	heavy	L2 R2 C3 D2 M0	yes
OTT subscriber app usage	Venn	Venn-regions+quant-DI	light	L1 R2 C1 D3 M0	partial
Country GDP per-capita	quant-DI	score-table	none	L0 R1 C0 D3 M1	no
AC temperature mode graph	chart-plus-logic	timeline	heavy	L2 R3 C3 D2 M3	yes
Foodgrain nutrient comp.	matrix/grid	matrix (incomplete)	heavy	L1 R2 C3 D2 M0	yes
10-number ordered grid	matrix/grid	grid (monotone)	heavy	L3 R3 C3 D0 M0	yes
Gated-area walkway routes	routes/network	flow-network	heavy	L1 R3 C2 D2 M0	partial
E-commerce daily ratings	chart-plus-logic	score-table	light	L1 R2 C1 D3 M0	partial
Firm PAT/ES/PRD bubble	chart-plus-logic	quant-DI	none	L0 R1 C0 D3 M1	no
Gymnasts coach ratings	distribution	allocation+score-table	heavy	L2 R3 C3 D1 M0	yes
QUIET tournament schedule	tournament	timeline	heavy	L2 R3 C3 D0 M0	yes
Travelers' Venn	Venn	Venn-regions (3-set)	light	L1 R2 C2 D2 M0	partial
Candlestick prices	share	chart-plus-logic	none	L0 R1 C0 D3 M1	no
Web surfers' star giving	distribution	allocation	heavy	L1 R3 C3 D2 M0	yes

Set		Family	Engine	Casework	Drivers	Ladder
Election voting	campaign	quant-DI	score-table+casework	heavy	L1 R2 C3 D3 M0	partial

16.9 CAT 2025 (15 sets — 5/slot; solution-calibrated; movement returns, ZERO Venn)

Set		Family	Engine	Casework	Drivers	Ladder
Operator minutes ledger	call-	matrix/grid	matrix+score-table	heavy	L3 R0 C2 D3 M0	partial
Cross-currency traveler costs	trav-	quant-DI	allocation+score-table	heavy	L2 R0 C2 D3 M0	yes
Puzzle-solving progress charts		chart-plus-logic	timeline (cross-chart)	light	L1 R0 C1 D3 M1	partial
Three/Five-country trade flows		quant-DI	flow-network	light	L1 R0 C1 D3 M0	yes
Passing-the-Buck circle		dynamic-movement	movement-traj.	heavy	L3 R2 C3 D0 M2	partial
Gurus training timeline		scheduling	timeline (blocks)	heavy	L3 R2 C3 D0 M0	partial
Sustainability-index scatter		chart-plus-logic	score-table+rank (rates)	heavy	L2 R2 C2 D3 M0	yes
State pollution measures		distribution	allocation+ranking	heavy	L3 R2 C2 D2 M0	partial
Co-authorship counts	paper	chart-plus-logic	matrix+allocation	heavy	L3 R1 C3 D2 M0	partial
Balls-through-hoops sizing		matrix/grid	matrix+ranking (order)	heavy	L3 R3 C2 D0 M0	yes
Train segment occupancy		distribution	flow-network (O-D)	heavy	L2 R1 C2 D3 M0	yes
Quarterly promote-demote	promote-	tournament	score-table+rank (fb)	heavy	L3 R2 C2 D2 M1	partial
Round-table moves	chair	dynamic-movement	movement-traj.	heavy	L3 R2 C3 D0 M3	partial
Foot-tap game	question	distribution	allocation+score-table	heavy	L3 R0 C3 D2 M0	partial
Five-country charts	tariff	quant-DI	flow-network	light	L1 R0 C1 D3 M0	yes

End of report.